

Pythagoras' Theorem & Trigonometry

Extra practice test 4 · 27 marks total

Name: _____ Date: _____ Score: _____ / 27

Section A — Pythagoras' theorem

A1. A right-angled triangle has two shorter sides both equal to 6 cm. Find the length of the hypotenuse, giving your answer both as a simplified surd and to 1 decimal place.

[2 marks]

A2. A right-angled triangle has a hypotenuse of 10 cm and one shorter side of 7 cm. Find the length of the other shorter side, to 1 decimal place.

[3 marks]

A3. A ladder reaches from the ground to a window 4.8 m up a wall. The foot of the ladder is 2.2 m from the wall. Find the length of the ladder, to 1 decimal place.

[2 marks]

Section B — SOH CAH TOA basics

B1. A right-angled triangle has a known hypotenuse and a known angle, and you need to find the side adjacent to that angle. Which method should you use — Pythagoras' theorem or SOH CAH TOA — and which specific ratio?

[1 mark]

B2. A right-angled triangle has two known shorter sides and no known angles. Which method should you use to find the hypotenuse: Pythagoras' theorem, or SOH CAH TOA?

[1 mark]

B3. True or False: Pythagoras' theorem can be used on any triangle, not just right-angled ones.

[1 mark]

Section C — Finding missing sides

C1. A right-angled triangle has an angle of 22° and an opposite side of 6 cm. Find the length of the hypotenuse, to 1 decimal place.

[3 marks]

C2. A right-angled triangle has an angle of 66° and a hypotenuse of 9 cm. Find the length of the side adjacent to the angle, to 1 decimal place.

[2 marks]

C3. A right-angled triangle has an angle of 39° and an opposite side of 7 cm. Find the length of the adjacent side, to 1 decimal place.

[2 marks]

Section D — Finding missing angles

D1. A right-angled triangle has a hypotenuse of 25 cm and an adjacent side of 7 cm. Find the angle between them, to 1 decimal place.

[2 marks]

D2. A zip-wire is fixed between two platforms. The horizontal distance between the platforms is 40 m and the vertical drop is 14 m. Find the angle the zip-wire makes with the horizontal, to 1 decimal place.

[3 marks]

Section E — Applying Pythagoras & trigonometry

E1. A right-angled triangle has an opposite side of 9 cm and an adjacent side of 12 cm. (a) Find the length of the hypotenuse. (b) Find the angle opposite the 9 cm side, to 1 decimal place.

[2 marks]

E2. A rectangular gate measures 1.8 m high and 3 m wide. A diagonal support strut is fitted. (a) Find the length of the strut, to 1 decimal place. (b) Find the angle the strut makes with the base of the gate, to 1 decimal place.

[3 marks]